Lab 03: Chaotic Text Generator & Sequence Tagging

Module: Natural Language Processing (NLP)

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1. The Problem and the Solution

Probabilistic text generation must balance syntactic regularity, lexical diversity and safety, yet it has no explicit ground truth. A standard first-order POS-HMM fails structurally because the Markov property prevents it from distinguishing subject nouns from object nouns. Higher-order n-gram baselines ensure local fluency but suffer from context amnesia and verbatim memorisation at low temperatures. We resolve the HMM's structural amnesia through syntactic state splitting, conditioning states on grammatical roles (Subject, Object, Preposition). This achieves 100% syntactic validity without iterative training, while Viterbi decoding enables reliable reverse sequence labelling.

2. Benchmark Metrics at T = 1.0 (500 sentences)

Model architecture	Valid tag pattern	Plausible subject-verb	Novel sentences	Distinct-2 (diversity)
HMM (plain states)	16.2%	14.8%	99.8%	0.198
HMM (role-split states: S / O / P)	100.0%	60.6%	99.4%	0.148
Trigram baseline	57.0%	76.0%	91.2%	0.105

The plain HMM reaches only 16.2% valid patterns because of the first-order Markov assumption: *P(tag_t | tag_t-1)* cannot track sentence progression. In the training data, after any *NOUN* the model moves to *VERB* (39%), *PREP* (22%) or *PUNCT* (39%), so it often produces ungrammatical output (e.g. ending the sentence right after a subject noun). The role-split HMM instead encodes position (*NOUN_S*, *NOUN_O*, *NOUN_P*): *NOUN_S* transitions to *VERB* with probability 1.0, which ensures a 100.0% valid tag pattern rate.

3. Temperature Selection for Specific Use Cases

Writing toy (T = 1.5): a higher temperature flattens the emission distributions and maximises lexical diversity (higher Distinct-2), encouraging playful and varied combinations despite occasional semantic quirks.

Autocomplete feature (T = 0.5): a low temperature concentrates probability on high-confidence, standard transitions, guaranteeing strict grammatical conformity and safe, predictable suggestions that do not disrupt the user's workflow.

4. Generation Failure Modes (Case Study)

Meaning failure: "the old birds sell at the river ." (role-split HMM, T = 1.0). The tag sequence is valid, but the emission probabilities *P(word | tag)* depend only on grammatical categories, with no world knowledge or selectional restrictions between words.

Memorisation failure: "the girls leave the park ." (trigram, T = 0.5). Low-temperature sampling follows the maximum-likelihood bigram/trigram contexts and reproduces training sentences verbatim instead of producing new text.

5. State Splitting: What It Changed vs. What It Did Not

State splitting gave the first-order HMM a structural memory by partitioning coarse states by clausal position (Subject, Object, Prepositional complement). This solved grammatical sequencing, raising valid patterns from 16.2% to 100%. However, it introduced no semantic awareness or lexical co-occurrence conditioning: words are still emitted independently of each other given the tag, so subject-verb plausibility stays well below the trigram baseline (60.6% vs. 76.0%).

6. Real-Data Treebank Evaluation (Universal Dependencies English EWT)

Evaluation regime / model	Accuracy	Known words	Unknown words (OOV)	Whole sentence correct
Synthetic corpus (HMM + Viterbi)	100.0%	100.0%	N/A (0.0% OOV)	100.0%
Real text: most-frequent-tag baseline	86.2%	90.8%	28.6%	31.9%
Real text: HMM + Viterbi	88.1%	92.9%	28.2%	40.4%

Viterbi reaches 100% accuracy on the synthetic data because the 61-word lexicon is unambiguous and only 16 regular patterns occur. On real web text, accuracy drops to 88.1% (only +1.9% over the unigram baseline). Real text shows high syntactic entropy (10,359 unique tag sequences) and 7.5% out-of-vocabulary tokens. For unknown words, the smoothed emissions favour rare tags with small denominators (e.g. *X*, *INTJ*), which shows that first-order sequence labelling alone cannot resolve open-domain lexical ambiguity without morphological subword features.